

Safe Multimodal Tumor Progression Forecasting via Mask Prediction from Longitudinal Brain MRI

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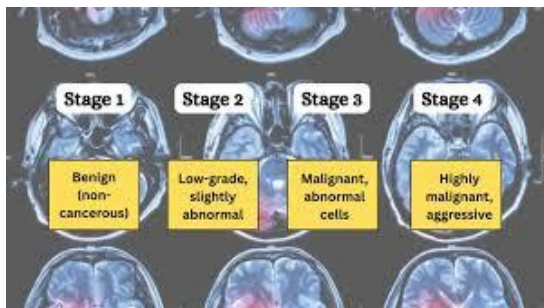
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1. Problem Statement

Brain tumor progression is commonly monitored using longitudinal magnetic resonance imaging (MRI). Accurate delineation of tumor regions across timepoints is essential for assessing disease evolution and treatment response. Deep learning approaches such as 3D U-Net and nnU-Net have demonstrated strong performance for static tumor segmentation tasks [1,2]. More recently, transformer-based architectures (e.g., UNETR) have improved volumetric representation learning for brain tumor analysis [3].

However, most existing methods focus on current-time segmentation and do not address the more clinically relevant problem of forecasting future tumor progression.

Longitudinal tumor response evaluation is typically performed retrospectively using criteria such as RANO [4], requiring multiple follow-up scans. Early prediction of tumor growth before the next imaging timepoint remains underexplored.



Furthermore, recent multimodal biomedical foundation models have shown that combining imaging and clinical text can enhance medical prediction tasks [5]. However, multimodal tumor progression forecasting that integrates MRI with clinical context remains limited. Additionally, safety concerns arise in medical AI systems. Studies have shown that deep neural networks can be poorly calibrated and overconfident in medical imaging tasks [6], which poses risks in clinical decision-making.

Therefore, we aim to address the following problem:

Given a baseline brain MRI and associated clinical text information, predict the future tumor segmentation mask at a subsequent time point while incorporating uncertainty-aware mechanisms for safe clinical deployment.

2. Motivation

Gliomas are highly heterogeneous and exhibit variable growth patterns depending on tumor subtype, molecular characteristics, and treatment interventions [4]. Accurate forecasting of tumor progression could assist in:

- Treatment planning
- Early identification of aggressive growth
- Personalized monitoring schedules
- Clinical risk stratification

Although segmentation benchmarks such as BraTS have advanced tumor delineation [7], the focus remains on static prediction. There is limited work on longitudinal forecasting of tumor evolution.

Moreover, multimodal learning approaches integrating imaging and clinical metadata have demonstrated improved performance in medical prediction tasks [5], suggesting that tumor progression modeling may benefit from joint MRI-text conditioning.

Finally, the importance of reliability in medical AI has been emphasized in recent literature, highlighting the need for calibrated uncertainty estimation and abstention mechanisms in safety-critical systems [6].

This project seeks to bridge these gaps by combining:

- Transformer-based volumetric feature extraction
- Clinical text conditioning
- Future mask prediction
- Uncertainty-aware selective prediction

3. Proposed Approach

Our proposed approach builds upon established 2D medical image segmentation frameworks and extends them toward **multimodal tumor severity prediction and longitudinal mask forecasting in 2D MRI slices**

Baseline

As a strong baseline, we adopt a 2D **U-Net architecture**, a widely used and well-established method for medical image segmentation. U-Net has demonstrated strong performance across tumor segmentation benchmarks and serves as a reliable starting point.

For multimodal severity prediction, a standard baseline consists of:

- Extracting image features from the encoder.
- Concatenating these features with structured clinical variables.
- Using a classification head to predict tumor severity.

For longitudinal modeling, a simple baseline approach predicts the future mask at time $t + \Delta$ using only image features extracted from time t , without incorporating clinical information or uncertainty modeling.

Proposal Method

We propose a **multimodal, safety-aware dual-task framework** operating on 2D MRI slices

1. **MRI Feature Encoding (ViT-Based 2D Backbone)**
A 2D Vision Transformer (ViT) encodes the MRI slice to capture global spatial context. The extracted features are used for segmentation, severity prediction, and future mask forecasting.
2. **Clinical Text Encoding (LLM-Based Representation)**
Clinical information (e.g., age, treatment, symptoms) is encoded using a pretrained clinical language model. This produces a patient-level embedding representing risk factors for tumor progression.
3. **Reliability-Gated Multimodal Fusion**
A learned gating mechanism adaptively weights image and text features. It detects modality conflicts and adjusts contributions based on confidence.
4. **Dual-Task Prediction**
 - **Tumor Severity Classification:**
Predicts tumor severity level.
 - **Future Mask Forecasting:**
Predicts the tumor segmentation mask at time $t + \Delta$, enabling quantitative tumor area estimation.

5. Safety-Aware Mechanisms

We integrate uncertainty estimation (Monte Carlo Dropout), confidence calibration, and selective prediction (abstention when uncertainty is high) to improve clinical reliability.

4. Prior Works

- **“Prediction of Brain MR Scans in Longitudinal Tumor Follow-up Studies” – Weizman et al., 2012.**
This work proposed a patient-specific model to predict future brain MRI scans in longitudinal tumor follow-up studies by estimating tumor growth parameters from prior scans. The method focused on forecasting volumetric tumor evolution but did not incorporate multimodal learning or uncertainty modeling.
- **“Longitudinal Brain Tumor Segmentation Prediction in MRI Using Feature and Label Fusion” – Pei et al., 2020.**
The authors introduced a framework to predict tumor segmentation in follow-up MRI scans using handcrafted features and label fusion strategies. Their approach modeled temporal tumor evolution but did not use transformer architectures or clinical text conditioning.
- **“Treatment-Aware Diffusion Probabilistic Model for Longitudinal MRI Generation and Diffuse Glioma Growth Prediction” – Liu et al., 2023.**
This study proposed a diffusion-based generative model to predict future MRI scans and tumor masks conditioned on treatment and time. While powerful, the approach focused on full image synthesis rather than safety-aware multimodal mask forecasting.
- **“DeepGrowth: Time-Conditioned Neural Fields for Tumor Growth Prediction” – Chen et al., 2024.**
The authors developed a time-conditioned neural network to model tumor growth dynamics from longitudinal MRI. Their framework predicted future tumor shapes using spatial-temporal modeling but did not incorporate multimodal clinical text or uncertainty-aware mechanisms.
- **“BioMedGPT: Open Multimodal Generative Pre-trained Transformer for Biomedicine” – Zhang et al., 2023.**

This work introduced a multimodal foundation model integrating medical images and biomedical text. Although not designed for tumor progression forecasting, it demonstrated the feasibility of combining visual and textual representations in medical AI systems, motivating our multimodal fusion strategy.

5. Plan Experiment

Baseline Model

Train a 2D image-only model to perform current tumor segmentation and severity prediction.

Multimodal Comparison

Compare image-only, naive fusion (MRI + text), and the proposed reliability-gated fusion for severity prediction.

Future Mask Forecasting

Predict the tumor mask at time $t+\Delta$ and evaluate using Dice score and area error.

Ablation Study

Evaluate the contribution of multimodal fusion and dual-task learning.

Evaluation

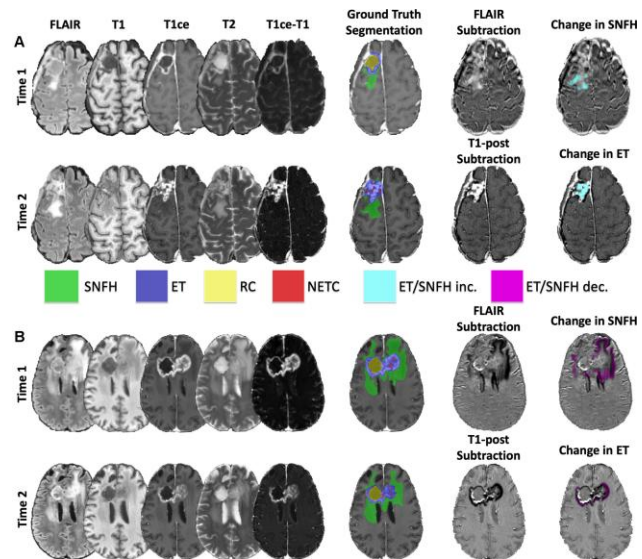
Measure calibration and evaluate selective prediction under uncertainty.

6. DataSets

We plan to use the UCSF Adult Longitudinal Post-Treatment Diffuse Glioma MRI Dataset, publicly available at:

<https://imagingdatasets.ucsf.edu/dataset/2>

This dataset contains longitudinal MRI scans from hundreds of adult diffuse glioma patients, including multiple follow-up timepoints per patient. It provides multimodal MRI sequences (such as T1, T2, FLAIR, and post-contrast T1), expert tumor segmentations of relevant subregions, and associated clinical metadata including treatment information. The longitudinal structure of the dataset enables supervised training for future tumor mask forecasting, while the availability of clinical variables supports multimodal modeling for tumor severity prediction.



References

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